Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 06)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 Table 1 shows the number of plant species per m^2 in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m^2	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

0 1 . 1 Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

.....

.....

.....

.....

0 1 . 2 Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 1 . 3 Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

.....

Do not write outside the box

Do not write outside the box

.....

.....

.....

0 1 . 4 Explain how burning maintains heathland.

[2 marks]

.....

.....

.....

.....

10

0 2 Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 2 . 1 Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	**Type of feedback**	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO_2 levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 2 . 2 Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

.....

.....

.....

.....

0 2 . 3 Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

.....

.....

.....

.....

0 2 . 4 Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

.....

.....

.....

.....

10

0 3 . 1 Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

.....

.....

.....

.....

0 3 . 2 For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

Do not write outside the box

Introduced competitor .....

.....

.....

.....

Introduced predator .....

.....

.....

.....

Introduced pathogen .....

.....

.....

.....

0 3 . 3 Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

.....

.....

.....

.....

10

0 4 . 1 Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

0 4 . 2 Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . 3 Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

Dissolved oxygen .....

.....

Water flow .....

.....

Day length .....

.....

Temperature .....

.....

10

0 5 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 5 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

Do not write outside the box

Do not write outside the box

.....

.....

.....

.....

0 5 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

.....

.....

.....

.....

5

0 6 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 6 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

.....

.....

.....

.....

0 6 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

.....

.....

.....

Do not write outside the box

5

0 7 A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 7 . 1 Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size

Distribution

Survival rate

Age structure

0 7 . 2 Describe how the population size of the antelope could be estimated.

[2 marks]

10

0 8 . 1 Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

0 8 . 2 Explain how pheromone traps can be used in pest control.

[2 marks]

0 8 . 3 Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

0 8 . 4 Give **one** way farmers can support crop pollinators.

[1 mark]

Do not write outside the box

Do not write outside the box

.....

.....

10

0 9 Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 9 . 1 Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer ..... %

0 9 . 2 Explain why so little of the solar energy is stored in the crop.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 9 . 3 Explain why few food chains have more than four trophic levels.

[2 marks]

.....

.....

Do not write outside the box

.....

.....

0 9 . 4 Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

.....

.....

.....

.....

0 9 . 5 Explain how pigs can increase the efficiency of a food production system.

[2 marks]

.....

.....

.....

.....

10

1 0 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

1 0 . 1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer ..... %

1 0 . 2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

Do not write outside the box

1 0 . 3 Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

1 0 . 4 Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

Do not write outside the box

Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

OR

1 1 . 2 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS